

Abstract

We develop and train a convolutional neural network (CNN) to classify short (5-second) audio clips into ambient environment categories. Raw audio sampled at 48 kHz is transformed into log-mel spectrograms, which are treated as single-channel images and fed into a four-layer 2D CNN with max-pooling, dropout regularization, adaptive average pooling, and a final fully connected classifier. The model is trained end-to-end using negative log-likelihood loss and the Adam optimizer, with an 80/20 split for internal validation. Our CNN model achieves 93.39% accuracy when classifying audio clips into five ambient sound categories.

Introduction & Background

Problem.

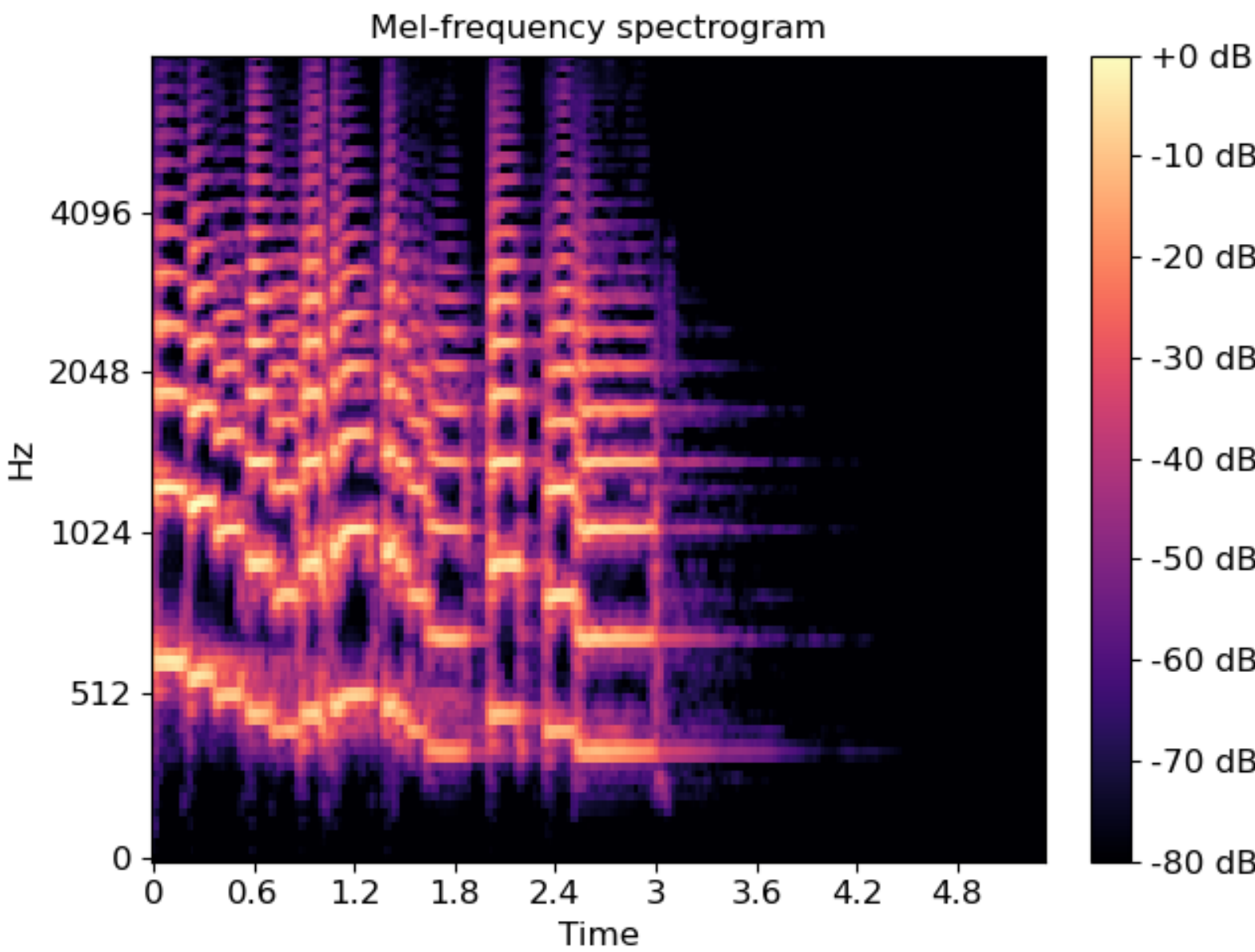
- Recognize ambient environments from short audio clips.
- Applications: context-aware phones, smart homes, environmental monitoring.
- Challenges: different devices, reverberation, overlapping sounds, limited labels.

Background.

- Environmental sound classification (ESC) commonly uses time–frequency features such as MFCCs and log-mel spectrograms, typically processed with 2D CNNs [1, 2].
- Benchmarks such as UrbanSound8K [3] and AudioSet [4] have established the use of these representations for ESC.

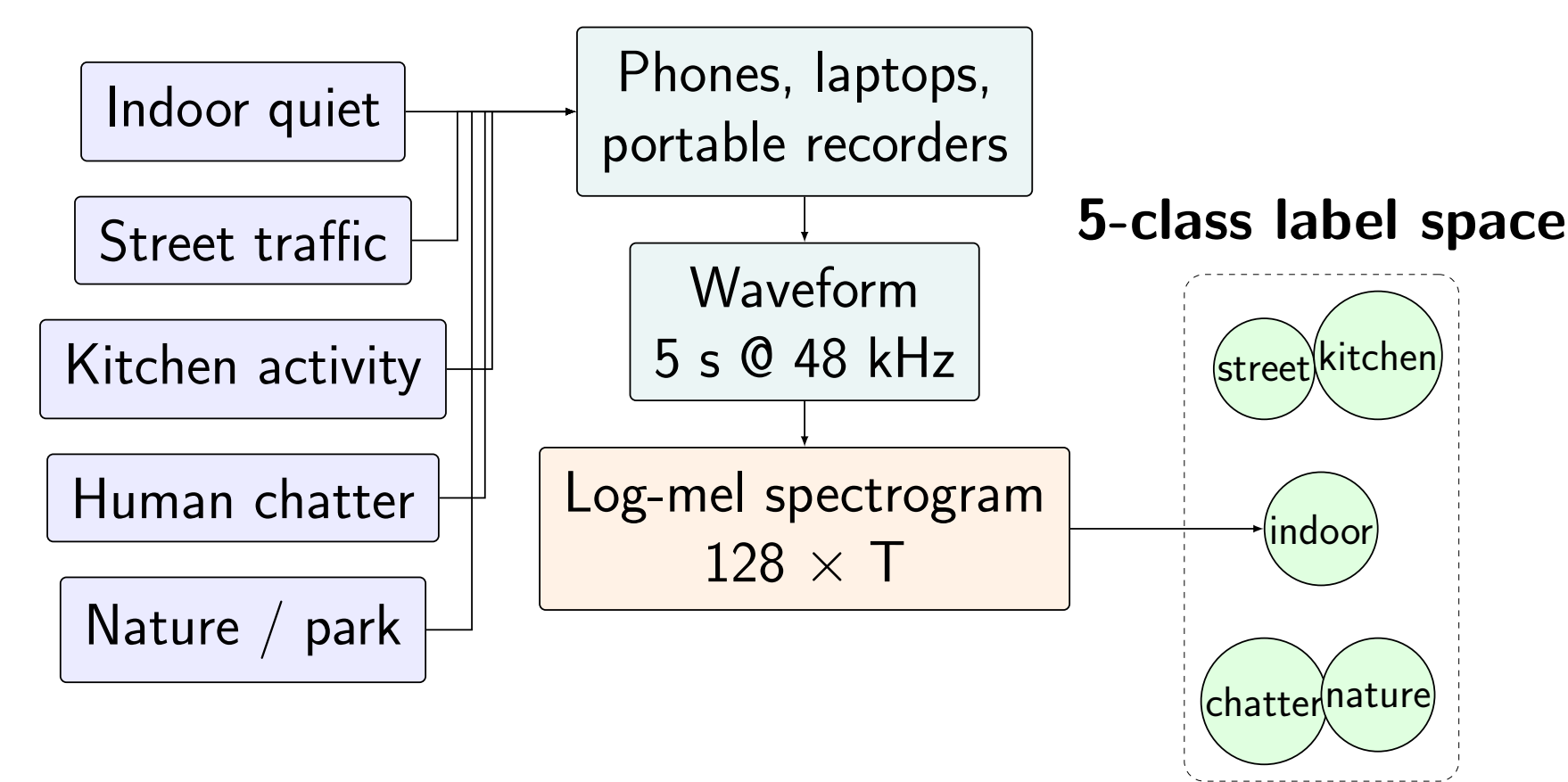
This Project.

- Dataset: 5 s, 48 kHz mono WAV clips in 5 ambient classes (indoor quiet, street traffic, kitchen activity, human chatter, nature).



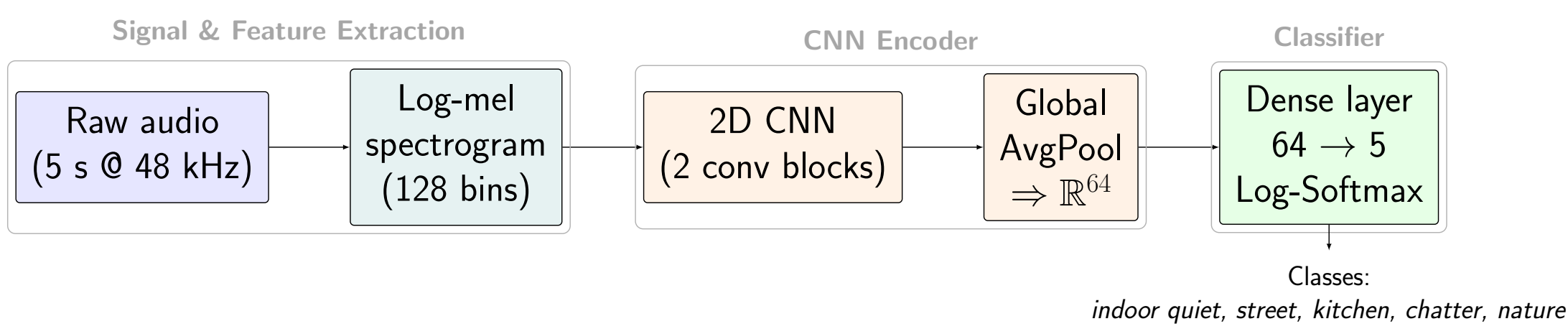
Example mel-spectrogram showing frequency content over time [5].

- Representation: 128-bin log-mel spectrograms fed to a compact 2D CNN.



Implementation

Model Architecture.



Algorithm 1 Ambient Sound Classification Pipeline (Training + Inference)

- Input:** Audio clip $x(t)$ (5 s, 48 kHz)
- Output:** Predicted class $p(y | x)$
- procedure** PREPROCESSING & AUGMENTATION
 - Normalize waveform amplitude
 - $S \leftarrow \text{MelSpec}(x)$ $\triangleright$ 128 mel bins
 - $\tilde{S} \leftarrow \text{power_to_dB}(S)$ $\triangleright$ Convert to log-mel
 - Apply data augmentation (pitch, time-stretch, noise)
 - Reshape to tensor $X \leftarrow [32, 1, 128, T]$
- end procedure**
- procedure** CNN ENCODER
 - $F \leftarrow \text{Conv2D} + \text{ReLU} + \text{Conv2D} + \text{ReLU}(X)$
 - $F \leftarrow \text{MaxPool2D}(F)$
 - $F \leftarrow \text{Dropout}(F)$
 - $F \leftarrow \text{Conv2D} + \text{ReLU} + \text{Conv2D} + \text{ReLU}(F)$
 - $F \leftarrow \text{MaxPool2D}(F)$
- end procedure**
- procedure** CLASSIFICATION
 - $z \leftarrow \text{AdaptiveAvgPool2D}(F)$ $\triangleright$ flatten to 64-D
 - $\ell \leftarrow Wz + b$ $\triangleright$ Linear layer
 - $p(y | x) \leftarrow \text{LogSoftmax}(\ell)$
- end procedure**
- procedure** TRAINING
 - for** each batch (X, y) in training set **do**
 - $p(y | X) \leftarrow \text{ForwardPass}(X)$
 - Compute loss: $L \leftarrow \text{NLLoss}(p(y | X), y)$
 - Backpropagate: $\nabla_{\theta} L$
 - Update weights: $\theta \leftarrow \theta - \eta \nabla_{\theta} L$ $\triangleright$ Adam optimizer
 - end for**
- end procedure**
- return** $p(y | x)$

Experiments & Results

Setup. 2D CNN on 128-bin log-mel spectrograms; we study the impact of data augmentation and regularization, with weighted F1 as the main metric.

Data Augmentation Pipeline.

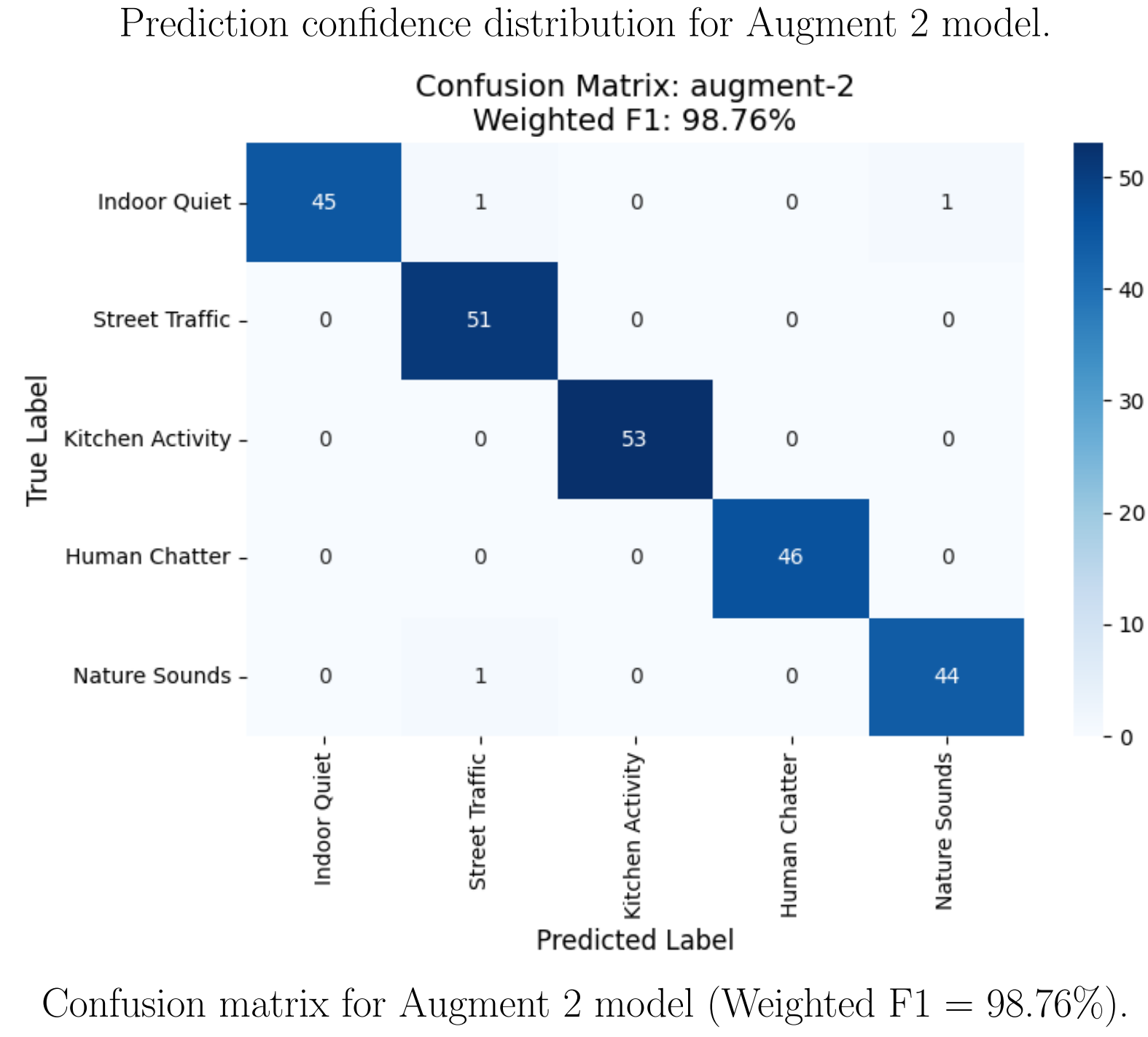
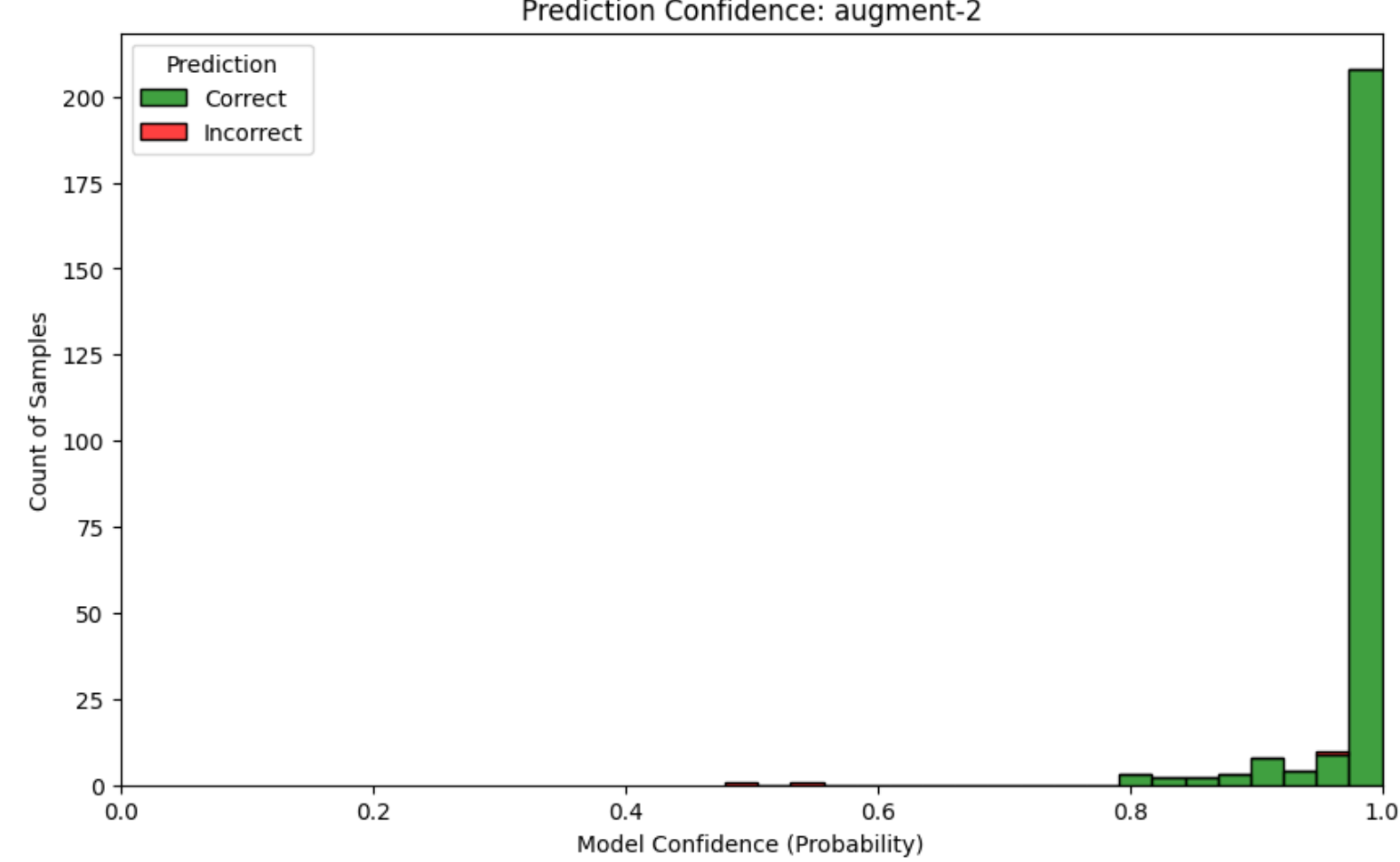
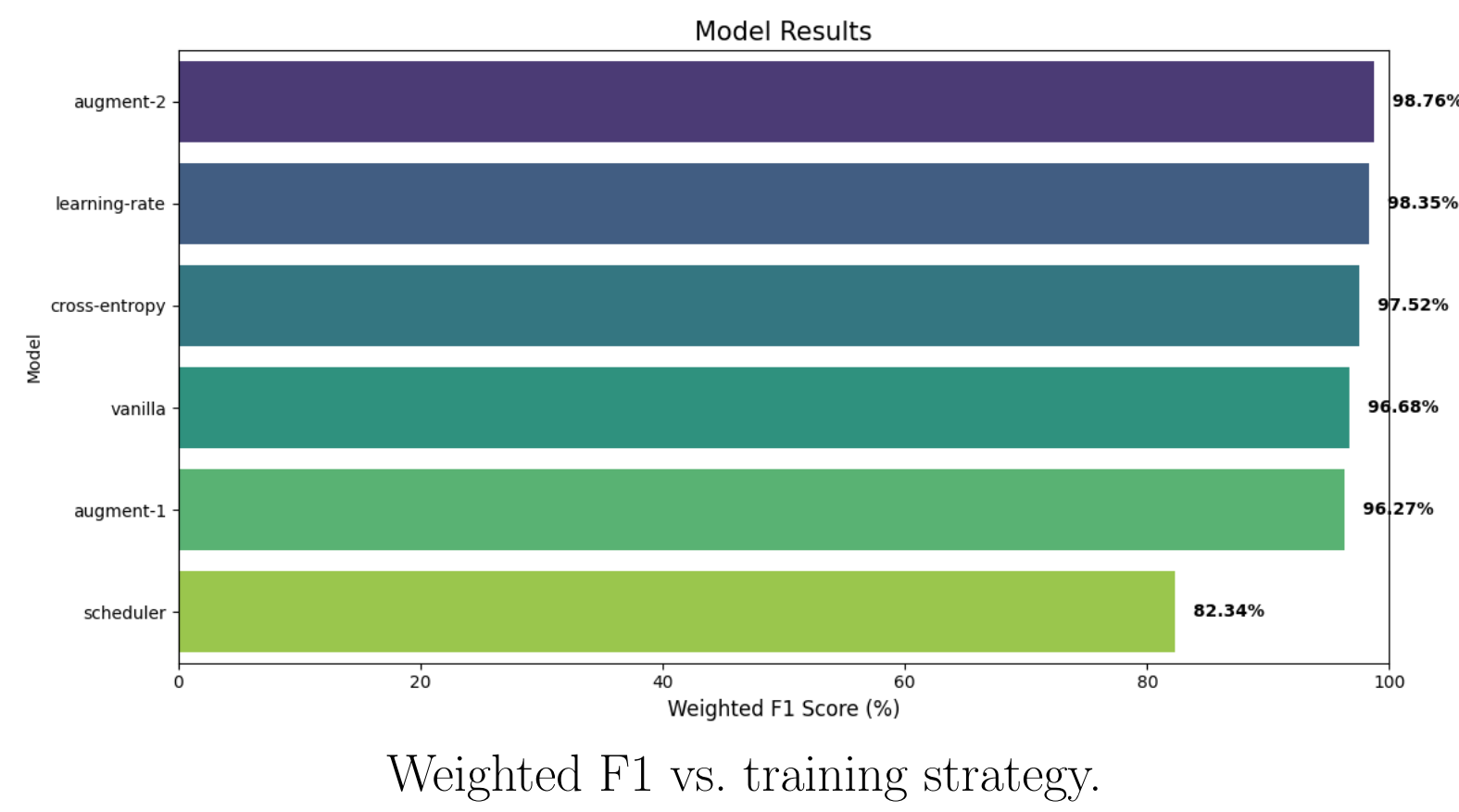
Technique	Type	Prob.	Parameters	Effect
Time Roll	Temporal	100%	0–100% shift	Different start times for continuous sounds.
Noise Injection	Signal	40%	0.005 factor	Simulate mic and background noise.
Time Stretch	Temporal	40%	0.85x–1.15x	Speed/duration invariance.
Pitch Shift	Spectral	40%	−2 to +1 steps	Pitch/timbre invariance.
Freq Masking	Spectral	50%	5–20 mel bins	Robust to missing frequency bands.

Training Strategies and Performance.

Model Trial	Key Strategy	Accuracy (training)	Outcome
Vanilla	Baseline (no augment)	86.78%	Overfits (high train, lower val).
Scheduler	Learning-rate decay	84.71%	LR drops early; undertrains.
Learning rate	Weight decay/L2 regularization	84.71%	Too constrained; underfits.
Cross-entropy	Label smoothing	90.08%	Softer decisions; modest gain.
Augment 1	Noise + stretch + pitch	93.39%	Breaks overfitting; clear boost.
Augment 2	+ time roll + masking	93.39%	Best generalization: peak F1.



Results & Discussion



Observations.

- Confusion matrix is nearly perfectly diagonal; only a few clips are misclassified.
- Residual errors occur between acoustically similar classes (e.g., quiet / nature vs. other low-energy environments); kitchen, chatter, and traffic are classified with no visible confusion.

Conclusions

- A compact 2D CNN on log-mel spectrograms achieves strong accuracy on 5 s ambient clips.
- Augmentations (pitch, time, noise, masking) greatly improve robustness and boost F1 to $\approx 99\%$.
- Performance gains come without increasing model size or dataset scale.
- Future work: broader ambient classes and open-set detection.

References

- [1] K. J. Piczak, “Environmental sound classification with convolutional neural networks,” in *Proceedings of the 25th International Workshop on Machine Learning for Signal Processing (MLSP)*, 2015.
- [2] S. Hershey, S. Chaudhuri, D. P. Ellis, J. F. Gemmeke, et al., “CNN architectures for large-scale audio classification,” in *ICASSP*, 2017.
- [3] J. Salamon, C. Jacoby, and J. P. Bello, “A dataset and taxonomy for urban sound research,” in *Proceedings of the 22nd ACM international conference on Multimedia*, 2014, pp. 1041–1044.
- [4] J. F. Gemmeke, D. P. Ellis, et al., “Audio set: An ontology and human-labeled dataset for audio events,” in *ICASSP*, 2017.
- [5] Librosa Developers, *librosa.feature.melspectrogram example figure*, Accessed: 2-Dec-2025. [Online]. Available: <https://librosa.org/doc/main/generated/librosa.feature.melspectrogram.html>.